

11. Early Pregnancy Loss: Diagnosing Nonviability Without Harm

Companion reading handout for simultaneous listening.

Companion reading handout for simultaneous listening.

Early pregnancy loss is common.

But the diagnostic error to fear is not that you fail to label every loss at the first visit.

The diagnostic error to fear is declaring a viable pregnancy nonviable.

The other error to fear is calling an empty uterus a miscarriage when the pregnancy may actually be ectopic.

So the dominant model for this lecture has two parts.

Certainty before evacuation.

Location before treatment.

Those two sentences are the safety architecture of early pregnancy diagnosis.

Early pregnancy loss is defined as a nonviable intrauterine pregnancy within the first twelve weeks and six days. It may be an empty gestational sac, or a gestational sac containing an embryo or fetus without cardiac activity.

The word intrauterine matters. If the pregnancy is not known to be intrauterine, you are not yet diagnosing early pregnancy loss. You are still working through pregnancy location.

Early pregnancy loss occurs in about ten percent of clinically recognized pregnancies.

About eighty percent of all pregnancy losses occur in the first trimester.

Approximately half are due to fetal chromosomal abnormalities.

Risk rises with maternal age and prior loss. ACOG cites clinically recognized loss rates of about nine to seventeen percent at ages twenty to thirty, twenty percent at age thirty five, forty percent at age forty, and eighty percent at age forty five.

Those numbers help with counseling. They also help residents avoid moralizing. Most early losses are biologic events, often chromosomal, not the result of routine activity, stress, exercise, or ordinary life.

The diagnostic problem

Bleeding and cramping can occur in a normal early pregnancy, early pregnancy loss, ectopic pregnancy, and molar pregnancy. Symptoms start the evaluation, but they do not settle it.

So begin with safety triage.

Is the patient hemodynamically stable? Is there syncope, dizziness, shoulder pain, severe unilateral pain, peritoneal signs, heavy bleeding, fever, or concern for infection? If she is unstable or has an acute abdomen, this is not a slow outpatient puzzle. Ectopic pregnancy, rupture, hemorrhage, or infection must be handled urgently.

If she is stable, move through the diagnostic sequence.

Confirm pregnancy

Second, locate the pregnancy.

Third, assess viability.

Fourth, decide whether the finding is diagnostic of nonviability or only suspicious.

That order prevents the most dangerous shortcuts.

History and examination still matter, even though ultrasound carries the diagnostic thresholds.

Ask about the amount of bleeding, clots, tissue passage, pain location, pain severity, dizziness, syncope, shoulder pain, fever, and prior ultrasound documentation. Ask the last menstrual period, but do not let the last menstrual period override a cautious scan. Ask about assisted reproduction, prior ectopic pregnancy, tubal surgery, pelvic infection, and pregnancy with an IUD in place, because those details raise ectopic concern.

On exam, decide whether she is stable, whether bleeding is excessive, whether there are peritoneal signs, whether the cervix is open, and whether there are signs of infection. A gentle pelvic exam can clarify bleeding source and cervical status, but it does not replace the location-and-viability framework.

The bedside question is not, "Does this look like miscarriage?" The bedside question is, "Can I prove what pregnancy this is, and can I prove whether it is viable?"

Ultrasonography, when available, is the preferred method to verify the presence of a viable intrauterine gestation. Serum hCG can help, but hCG alone cannot diagnose location or viability.

Now pause on why the ultrasound criteria are conservative.

Earlier practice used smaller cutoffs. A crown-rump length of five millimeters without cardiac activity, or an empty gestational sac with mean sac diameter of sixteen

millimeters, were once used to diagnose early pregnancy loss.

But prospective studies challenged those older cutoffs. ACOG summarizes data in which a five millimeter crown-rump length cutoff without cardiac activity had an eight point three percent false-positive rate for early pregnancy loss. The mean sac diameter cutoff of sixteen millimeters had a four point four percent false-positive rate. To reach zero false positives in that study, the crown-rump length cutoff needed to be five point three millimeters, and the mean sac diameter needed to be twenty one millimeters.

The modern consensus thresholds are even more conservative.

Why? Because the cost of a false positive is catastrophic. A false positive does not just create a wrong label. It can lead to uterine evacuation or medication treatment of a pregnancy that might have continued.

This is one of the places where residents should become comfortable with deliberate slowness. In many areas of medicine, speed is safety. In this narrow diagnostic space, when the patient is stable and the finding is borderline, speed can become harm.

The embryo may be only a few millimeters long. The sac measurement may vary between observers. Dating may be wrong. Ovulation may have been later than expected. A scan done today may sit just before the moment when a yolk sac, embryo, or heartbeat would have appeared.

That is why the diagnostic lines are set beyond the gray zone.

This is why the numbers are not arbitrary.

They are safety boundaries.

The diagnostic ultrasound criteria

A crown-rump length of seven millimeters or greater with no heartbeat is diagnostic of pregnancy failure.

A mean sac diameter of twenty five millimeters or greater with no embryo is diagnostic.

No embryo with heartbeat two weeks or more after a scan that showed a gestational sac without a yolk sac is diagnostic.

No embryo with heartbeat eleven days or more after a scan that showed a gestational sac with a yolk sac is diagnostic.

These are the lines where the diagnosis becomes secure enough to act, assuming the clinical picture fits and the pregnancy is intrauterine.

Contrast that with suspicious findings

A crown-rump length less than seven millimeters with no heartbeat is suspicious, not diagnostic.

A mean sac diameter sixteen to twenty four millimeters with no embryo is suspicious, not diagnostic.

Absence of an embryo with heartbeat seven to thirteen days after a scan that showed a gestational sac without a yolk sac is suspicious.

Absence of an embryo with heartbeat seven to ten days after a scan that showed a gestational sac with a yolk sac is suspicious.

No embryo six weeks or longer after the last menstrual period is suspicious.

An empty amnion is suspicious.

A yolk sac greater than seven millimeters is suspicious.

A small gestational sac relative to the embryo, with less than five millimeters difference between mean sac diameter and crown-rump length, is suspicious.

Slow fetal heart rate, less than one hundred beats per minute at five to seven weeks, and subchorionic hemorrhage are also associated with early pregnancy loss. But they are not diagnostic.

The resident reflex is this: suspicious is not diagnostic.

Suspicious findings generally warrant follow-up ultrasound in seven to ten days. That follow-up is not procrastination. It is how the diagnosis becomes safe.

Of course, the patient is part of the diagnostic process. ACOG emphasizes individualization. Consider whether she desires to continue the pregnancy if viable, how much certainty she wants before intervention, the consequences of waiting, the possibility of spontaneous tissue passage, the risk of an unscheduled visit, and the anxiety caused by uncertainty.

But individualization does not mean pretending uncertain findings are certain. It means explaining the uncertainty honestly and choosing a plan that respects both safety and the patient's situation.

There is a professional way to say this at the bedside.

"I am concerned this pregnancy may not be developing normally. But today's scan does not meet the criteria that let us diagnose loss with enough certainty. If we act today, we could be wrong. The safer plan is to repeat the scan after enough time has passed for the pregnancy to declare itself."

That kind of language validates the concern without overstating the diagnosis.

hCG is useful, but it is not a map.

A single hCG value cannot diagnose viability or location. It can support follow-up planning, especially when ultrasound is nondiagnostic, but it must be interpreted with symptoms, exam, ultrasound, and time.

This point becomes critical when no intrauterine pregnancy is seen.

A pregnant patient without definitive intrauterine or ectopic pregnancy on ultrasound has a pregnancy of unknown location. That is not a final diagnosis. It is a transient state.

If she is clinically stable and would want to continue the pregnancy if it is intrauterine, she should have repeat transvaginal ultrasound, serial hCG, or both, to confirm the diagnosis and guide management.

Do not give methotrexate for a pregnancy of unknown location just because you are tired of uncertainty. Methotrexate can interrupt or harm an ongoing intrauterine pregnancy. That error is exactly what the diagnostic framework is built to prevent.

The minimum evaluation for suspected ectopic pregnancy is confirmation of pregnancy and transvaginal ultrasound. Serial ultrasound, serial hCG, or both may be required. If the patient is unstable or has an acute abdomen, urgent evaluation and treatment take priority.

Definitive ultrasound evidence of an intrauterine pregnancy is a gestational sac with a yolk sac or embryo. A sac-like fluid collection alone can be misleading, because a pseudogestational sac can occur with ectopic pregnancy. Once a definitive intrauterine pregnancy is seen, ectopic pregnancy is generally excluded, except for the rare possibility of heterotopic pregnancy, which matters more in assisted reproduction.

The discriminatory level

The discriminatory level is the hCG value above which landmarks of a normal intrauterine gestation should be visible on ultrasound. It is tempting to use it as a bright line. But ACOG cautions that the discriminatory level can mislead. There are cases where an intrauterine gestational sac was seen on follow-up even though no sac was seen initially and hCG was above the discriminatory level.

If the discriminatory concept is used, the value should be conservatively high, as high as three thousand five hundred milli international units per milliliter, to avoid misdiagnosis and possible interruption of an intrauterine pregnancy the patient hopes to continue.

Multiple gestations can also have higher hCG values before ultrasound landmarks are visible.

So again, hCG is not a map.

It is a signal from trophoblast. It does not tell you by itself whether the pregnancy is intrauterine, ectopic, viable, or nonviable.

Serial hCG trends help, but even trends have limits. A rising hCG can still be abnormal. A falling hCG can occur with a resolving intrauterine loss, but ectopic pregnancy can rupture even with low or declining values. That is why follow-up continues until the diagnosis is secure or the pregnancy resolves safely.

Use the model in cases

Case one.

A patient has bleeding. Ultrasound shows an embryo with crown-rump length six millimeters and no heartbeat.

This is concerning, but it is not diagnostic. The correct next step, if she is stable, is follow-up ultrasound, generally in seven to ten days. Do not treat this as confirmed loss.

Case two.

The mean sac diameter is twenty two millimeters, with no embryo.

Again, suspicious, not diagnostic. The diagnostic threshold is twenty five millimeters or greater.

Case three.

An ultrasound shows a gestational sac without a yolk sac. Seven days later, there is still no embryo with heartbeat.

This is still not the two-week diagnostic interval. It is suspicious. Continued follow-up may be needed if the patient desires certainty before intervention and remains stable.

Case four.

A scan shows a gestational sac with a yolk sac. Eleven days later, there is no embryo with heartbeat.

That meets a diagnostic criterion for pregnancy failure.

Case five.

A patient has a positive pregnancy test, bleeding, and no intrauterine pregnancy on ultrasound.

Do not call this miscarriage. The correct label is pregnancy of unknown location until further evaluation establishes intrauterine loss, ectopic pregnancy, or another outcome. Serial hCG and repeat ultrasound are often required.

If the patient wants the pregnancy if it is intrauterine, this distinction becomes even more important. The diagnostic plan must protect her from both risks: untreated ectopic pregnancy and accidental interruption of a desired intrauterine pregnancy.

Case six.

A patient had a documented intrauterine pregnancy on prior ultrasound. She now presents after significant bleeding, and ultrasound shows an empty uterus.

That can be a straightforward diagnosis of early pregnancy loss, because the location was previously documented. This is very different from an empty uterus in a patient whose pregnancy location was never confirmed.

Case seven.

The scan shows a fetal heart rate of ninety beats per minute at six weeks.

That is worrisome. The risk of subsequent loss is higher. But it is not a diagnosis of pregnancy failure. The correct language is guarded prognosis and follow-up, not treatment as a confirmed loss.

Case eight.

The scan shows a subchorionic hematoma in a patient with bleeding.

Again, this can be associated with loss, but it is not diagnostic of loss. The patient needs counseling and follow-up, not a premature declaration that the pregnancy is over.

Counseling during uncertainty

Tell the patient what you know, what you do not yet know, and what finding would change the plan.

For example: "Right now the ultrasound is concerning, but it does not meet the criteria that allow us to diagnose pregnancy loss safely. The reason we repeat the scan is to avoid treating a pregnancy that could still be viable."

Also give clear return precautions: heavy bleeding, dizziness, fainting, worsening abdominal or shoulder pain, fever, or feeling acutely unwell. The patient should understand that uncertainty is being monitored, not ignored.

The final synthesis

Early pregnancy loss diagnosis is a safety discipline.

First: is she stable?

Second: where is the pregnancy?

Third: is it viable?

Fourth: are the ultrasound findings diagnostic, or only suspicious?

Diagnostic means crown-rump length seven millimeters or greater without heartbeat, mean sac diameter twenty five millimeters or greater without embryo, no heartbeat two weeks or more after a sac without yolk sac, or no heartbeat eleven days or more after a sac with yolk sac.

Suspicious findings require follow-up, usually seven to ten days, not premature treatment.

hCG helps with follow-up, but hCG is not a map. Pregnancy of unknown location is not early pregnancy loss until location and viability are resolved.

And the final resident reflex is this.

In a stable patient, time and repeat imaging can be safer than premature certainty.